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Augmented Auditory Reality for Noise Reduction in Hearing Aids

Abstract

User-intent information can support speech enhancement by allowing hearing systems to prioritize sources according to listener behavior. Electroencephalography (EEG) and electrooculography (EOG) have been investigated for this purpose, but their requirements complicate integration into compact hearing devices, whereas head orientation provides a simpler interaction cue. This work presents a five-experiment benchmark for head-orientation-driven Augmented Auditory Reality (AAR) speech enhancement using the Tiresias research platform, a rotating acoustic rig, measured two-channel responses, and 100 LibriSpeech source pairs. The experiments characterized orientation and Bluetooth Low Energy (BLE) telemetry, acoustic-response reconstruction, Gaussian attention width, control delay, source leakage, and source-estimate delay. The BMI270 yaw estimate drifted between 2.44 and 3.23°/min, while post-hoc trend removal reduced the residual Root Mean Square Error (RMSE) to 0.47°. BLE notifications were received at 40.7 Hz, and cross-repetition reconstruction achieved a prediction Signal-to-Distortion Ratio of 23.23 dB and a correlation of 0.997. For sources at $\pm 30°$, $\sigma = 30°$ produced the highest mean Target-to-Interferer Ratio improvement, at 5.47 dB. At this width, control delays up to 80 ms caused at most 1.35 dB of loss. In contrast, source leakage substantially reduced the enhancement, and source-estimate delays of 20 ms or more exceeded the adopted intelligibility-loss criterion. Practical implementation therefore requires absolute heading correction and an attention width matched to source geometry, while low-latency, low-leakage source separation remains the critical system requirement.

Keywords: Augmented Auditory Reality, head orientation, hearing assistance, source separation, speech enhancement.

1. Introduction

Understanding speech in the presence of competing sources remains difficult for hearing-assistance systems. Current hearing aids combine amplification, dynamic-range compression, directional processing, and speech enhancement, yet their behavior in everyday situations can differ from laboratory results because listeners and talkers move and the acoustic scene changes continuously [1–3]. This has motivated hearing technologies and experimental methods that consider listener behavior, device processing, and the acoustic environment as interacting parts of the same system [3, 4]. In this context, listener motion can be used as input to the signal processing.

Augmented Auditory Reality (AAR) provides one framework for this interaction. Acoustic sources can be represented as Virtual Sound Objects (VSOs), each containing its own spatial metadata and control parameters [5]. When source separation provides individual source estimates, these objects can be processed independently and later combined into a new enhanced acoustic scene, allowing behavioral cues to determine which source receives additional gain. Grimm et al. investigated gaze-guided source selection and reported benefits comparable to conventional directional processing under the conditions evaluated in their study [6]. For hearing assistance, this approach suggests a route to selective enhancement without completely removing information from the surrounding environment.

The selection cue can be obtained from different sensing modalities. Auditory-attention decoding has been investigated using electroencephalography (EEG), while electrooculography (EOG) and related ocular signals can provide information about eye movements and visual orientation [7, 8]. These approaches usually require additional sensing hardware, electrode placement, and signal processing that may complicate or render integration and embedding on compact hearing devices infeasible. Head movements have been associated with source selection in conversational scenes [9, 10], making it a logical candidate for a lower-complexity alternative since it can be measured with small, low-power inertial sensors.

Object-level enhancement requires source estimates that remain sufficiently distinct, synchronized, and associated with the correct positions in the acoustic scene, and angular orientation estimation poses the main challenges for these types of systems and frameworks, where the orientation signal must be sufficiently accurate and timely, while the angular width of the gain function must balance source differentiation against tolerance to head motion.

This work develops and evaluates an experimental benchmark for head-orientation-driven AAR speech enhancement, structured as five complementary experiments that characterize both the orientation-based control path and the separated-source audio path. Using the Tiresias hearing research platform [11], a custom rotating acoustic rig, measured two-channel acoustic responses, and controlled offline perturbations, the benchmark evaluates orientation accuracy and update timing, acoustic-response reconstruction, Gaussian attention width, control delay, source leakage, and source-estimate delay.

The main contributions are: (i) an experimental benchmark linking Tiresias inertial and Bluetooth Low Energy (BLE) telemetry to an object-level AAR speech-enhancement model; (ii) a full-rotation characterization of the orientation-control path and a cross-repetition evaluation of measured two-channel acoustic responses; (iii) controlled sensitivity analyses of Gaussian attention width, orientation-control delay, source leakage, and source-estimate delay; and (iv) a preliminary set of engineering requirements for the orientation-sensing and source-separation subsystems of a practical implementation.

The remainder of this paper is organized as follows. Section 2 describes the head-orientation-based attention model and the measured two-channel rendering procedure. Section 3 presents the orientation, acoustic-measurement, and offline-benchmark protocols. Section 4 reports the results for orientation telemetry, BLE update timing, acoustic reconstruction, attention width, control delay, and source-separation impairments. Section 5 discusses the main engineering implications and limitations, and Section 6 concludes the paper.

2. Head-Orientation-Driven AAR Model

Figure 1 presents the signal flow considered in the benchmark. The control path begins with inertial measurements acquired by the IMU on the Tiresias platform. The estimated head orientation is transmitted to the attention model, which determines the gain assigned to each source object. In parallel, the audio path assumes that the binaural microphone signals are processed by a source-separation stage, producing two individually controllable source estimates, denoted as sources A and B. The attention model applies the orientation-dependent gains to these estimates, generating the signal delivered to the hearing device output and used in the objective evaluation.

The benchmark considers two monophonic source objects positioned at azimuths $-30°$ and $+30°$ with respect to the frontal acoustic reference. Source separation precedes the attention stage by calculating the scalar gain for each source. The same gain is applied to its left and right acoustic images.

For a given listener orientation, represented by a unit quaternion $\mathbf{q} = [q_w, q_x, q_y, q_z]$, a forward-looking unit vector $\vec{v}_{\text{forward}}$ is computed. The angular separation between the head-forward direction and source s is then

$$\alpha_s = \arccos\left[\text{clamp}\left(\vec{v}_{\text{forward}} \cdot \vec{u}_s, -1, 1\right)\right], \quad (1)$$

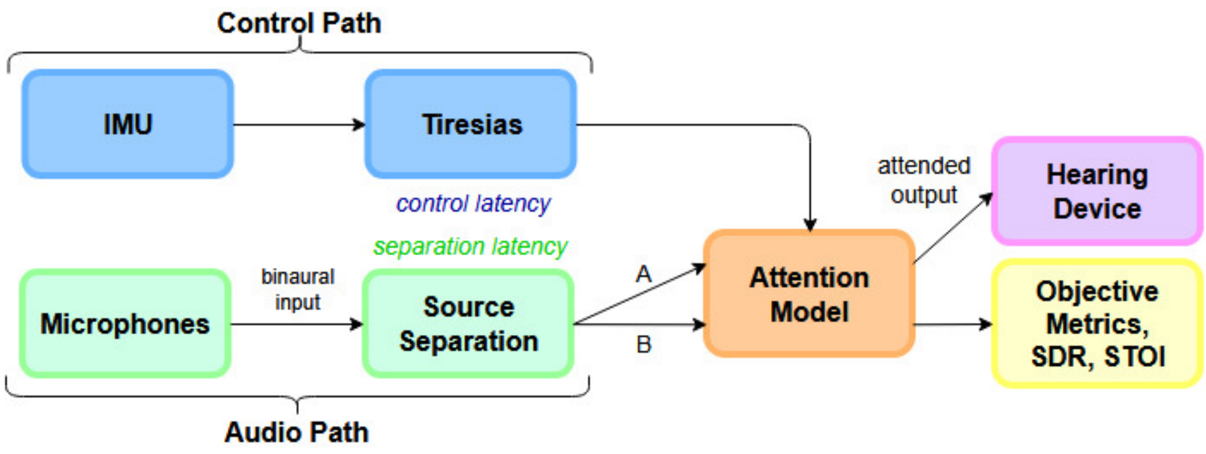


Figure 1: Conceptual block diagram of the evaluated AAR benchmark.

where $\vec{u}_s$ is the unit vector pointing toward the source.

A Gaussian angular profile was adopted to reproduce, at the source-object level, the smooth directional selectivity associated with a beamformer. Instead of changing abruptly between sources, the gain varies continuously as the head rotates. The attention weight and corresponding gain are given by

$$A_s = \exp\left(-\frac{\alpha_s^2}{2\sigma^2}\right), \qquad g_s = 10^{B_{\max} A_s/20}, \tag{2}$$

When the head-forward direction is aligned with source s, $\alpha_s = 0$, resulting in $A_s = 1$ and a gain of $B_{\max}$. As the angular distance from the source increases, the applied gain decreases continuously toward 0 dB. Smaller values of σ generate a narrower directional pattern, whereas larger values distribute the gain over a wider angular region.

Figure 2 shows the gain patterns used for the nominal condition $\sigma = 30^\circ$. The gain assigned to source A reaches its maximum when the listener faces -30°, while the gain assigned to source B reaches its maximum at $+30^\circ$. Between these directions, the gains vary continuously, allowing the dominant source to change gradually as the head rotates.

To include the source distance in the rendering model, the angular gain is combined with an inverse-distance factor,

$$g_{\text{final},s} = G_{\text{dist},s}\, g_s, \qquad G_{\text{dist},s} = \frac{d_{\text{ref}}}{\max(d_s, d_{\text{ref}})}, \tag{3}$$

where d_s is the distance between the listener reference point and source s, and $d_{\text{ref}} = 1$ m. In the present benchmark, both loudspeakers were positioned at the reference distance, so the two sources had the same distance factor.

The attention model calculates one time-varying scalar gain for each source object. The same gain is applied to the left and right components of that source, and the output channels are obtained by summing the gain-adjusted objects:

$$y_e[n] = g_A[n]\hat{x}_{A,e}[n] + g_B[n]\hat{x}_{B,e}[n], \qquad e \in \{L, R\}. \tag{4}$$

Here, $\hat{x}_{s,e}[n]$ denotes the component of source s in channel e. Consequently, head orientation controls the relative level of the source objects, while the existing relationship between the left and right components of each object is preserved.

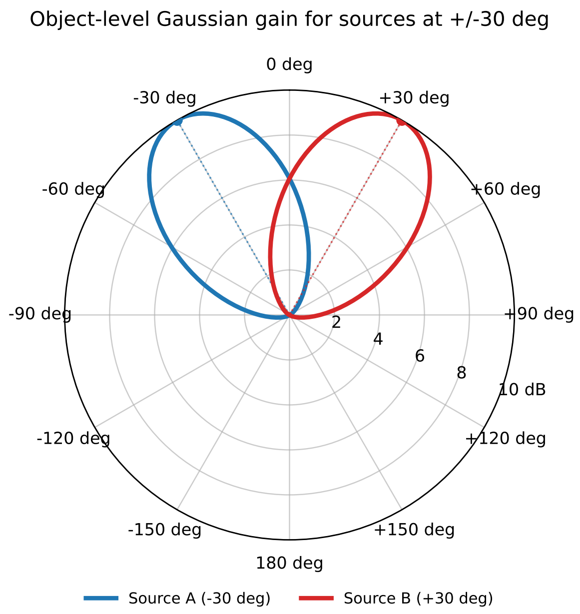


Figure 2: Object-level Gaussian gain patterns for sources A and B, positioned at -30° and $+30^\circ$, respectively. The curves were generated using $\sigma = 30^\circ$ and $B_{\max} = 10$ dB and show the gain applied to each source as a function of head yaw.

3. Experimental Framework

Five experiments were designed to characterize the two main paths of the proposed system. Experiment I evaluated the orientation signal used by the attention controller, including angular error, yaw drift, and BLE update timing. Experiment II measured and evaluated the orientation-dependent two-channel acoustic responses used to represent the physical scene. Experiments III–V then used these responses in an offline speech benchmark to study the Gaussian attention width, orientation-control delay, source leakage, and source-estimate delay.

3.1 Experimental Setup and Physical Measurements

The physical setup is shown in Fig. 3. A custom rotating mannequin rig maintained the Tiresias board and the two microphones as a rigid assembly. The platform consisted of a fixed 60 cm MDF disk and a rotating 40 cm disk connected by a central pivot. Angular marks were drawn every 5° from 0° to 360°, and measurements were performed every 10°.

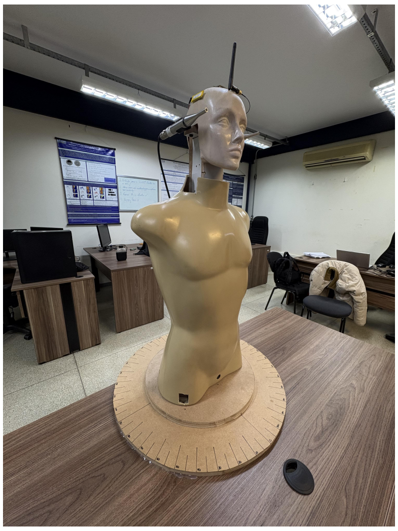
In Experiment I, the BMI270 measurements acquired by Tiresias were converted to orientation quaternions and transmitted to a host application over Bluetooth Low Energy (BLE). Physical reference angles ranged from 0° to 360° in 10° increments. The final 360° position represented a return to the initial orientation and was used as a closure measurement rather than as an additional direction in the global statistics.

Three runs were recorded in ascending, descending, and randomized angular order. At each position, the protocol included 3 s for settling and 10 s of acquisition. The first 2 s were discarded, leaving an 8 s analysis interval. Circular angular error was computed as

$$\theta_h - \theta_{\text{ref}}, \tag{5}$$

where θ_{ref} is the platform angle and θ_h is the yaw obtained from the received quaternion. The analysis included angular error, closure, drift, received-notification intervals, update rate, and packet continuity.

For Experiment II, two Neumann KH80DSP monitors were positioned at -30° and $+30^\circ$, each 1.0 m from the mannequin frontal reference. Two Earthworks M23R microphones were mounted at the left and



(a) Rotating mannequin rig, angular base, Tiresias board, and Earthworks M23R microphones.

(b) Complete setup with two Neumann KH80DSP monitors, a Focusrite Scarlett 18i8 interface, and the acquisition computer.

Figure 3: Experimental setup used for the orientation and acoustic-response measurements.

right sides of the head. A Focusrite Scarlett 18i8 interface simultaneously recorded the two microphone signals and an electrical-reference channel at 48 kHz, and the microphone factory magnitude-calibration files were applied during processing.

The platform was rotated clockwise through 37 positions, $0, 10, \ldots, 350, 360^\circ$. Both loudspeakers were measured independently at each angle, with two repetitions per condition. For each loudspeaker, orientation, and repetition, the left and right impulse responses form an orientation-dependent two-channel acoustic response, referred to hereafter as a binaural room impulse response (BRIR). This procedure produced 148 BRIR measurements, comprising 296 single-channel impulse responses. The responses were evaluated by reconvolution: the recorded electrical reference was convolved with an estimated response and compared with the corresponding microphone signal. Same-trial evaluation used the same sweep for estimation and reconstruction, whereas cross-repetition evaluation used one repetition to predict the other. Prediction SDR, waveform correlation, and normalized root-mean-square error were calculated.

3.2 Offline Benchmark Conditions

Experiments III–V used 100 non-overlapping speech pairs selected from a 200-file LibriSpeech *dev-clean* subset using seed 2026 [12]. Each segment was 6 s long. Sources A and B were assigned to the measured loudspeaker positions at -30° and $+30^\circ$, and the same speech pairs, source assignments, and acoustic responses were used across all parameter conditions.

Experiment III evaluated $\sigma \in \{10, 20, 30, 45, 60\}^\circ$ for head orientations between -90° and $+90^\circ$. For each target source, the improvement in Target-to-Interferer Ratio (ΔTIR) was averaged over the angular region in which that source was no farther from the head direction than the interferer. The result was normalized by angular span, averaged over both target assignments, and summarized with a percentile bootstrap over the 100 speech pairs. The lower bound of the 95% bootstrap interval was used to rank the tested attention widths.

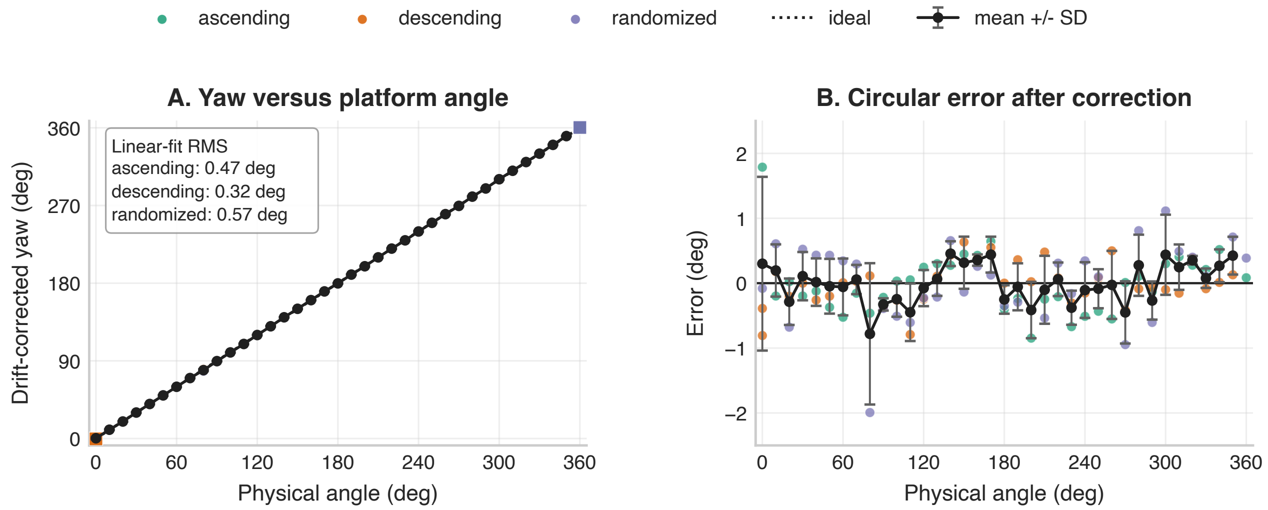


Figure 4: Measured orientation telemetry from the Tiresias platform over a full physical revolution.

Experiment IV evaluated orientation-control delays $\tau_c \in \{0, 10, 20, 40, 80, 120, 160, 200\}$ ms at angular velocities of 30, 60, and 120 °/s. The acoustic trajectory followed the current head orientation, while the gains were calculated from delayed yaw samples. TIR loss relative to the corresponding zero-delay condition and RMS source-gain error were used to quantify the effect of the delay.

Experiment V evaluated symmetric cross-source leakage using nominal separator Signal-to-Distortion Ratio values of $\{\infty, 20, 10, 5, 0\}$ dB and source-estimate delays of $\{0, 20, 40, 80, 120, 160, 200\}$ ms, with orientation-control delay fixed at 0 ms. TIR improvement was computed from the preserved target and interference components, and Short-Time Objective Intelligibility (STOI) [13] was also compared with the zero-delay, no-leakage reference condition.

4. Results

4.1 Experiment I - Orientation and BLE Telemetry

The orientation measurements are summarized in Fig. 4, where the figure compares physical platform angle and calibrated yaw using circular-angle processing. The 360° endpoint is retained as a closure measurement. Before drift correction, monotonic drift dominated the error, producing a Mean Absolute Error (MAE) of 20.43°and a Root Mean Square Error (RMSE) of 23.74°. Applying a linear fit over the angular data gave a drift slope of -3.23 °/min for the worst case. After post-hoc linear correction, the global MAE was reduced to 0.36°, the RMSE to 0.47°, and the maximum absolute error to 2.04°.

The local repeatability of the inertial yaw estimate was therefore much better than its long-term stability. Relative to the selected 30° Gaussian width, the corrected sub-degree RMSE is small, but the uncorrected drift accumulated over several minutes would dominate the orientation value. This behavior is consistent with a yaw estimate obtained without magnetometer information, and a practical implementation would consequently require an IMU with a magnetometer for online yaw correction.

BLE timing was comparatively stable over the three runs of experiment I, with an average notification rate of 40.7 Hz, a 95% confidence interval of 40.7 ms, and a 99% confidence interval of 59.4 ms. Packet loss, measured relative to the modal sequence-number increment, averaged 0.11%. Table 1 and Fig. 5 show the observed update jitter. These values describe notification spacing and continuity only and are not measurements of absolute sensor-to-host latency since no independent synchronized physical clock was available.

Table 1: BLE notification timing during Experiment I. Packet loss is computed relative to the modal sequence-number increment.

Run	Rate (Hz)	P95 interval (ms)	P95 jitter (ms)	Loss (%)
ascending	40.8	37.5	17.6	0.00
descending	40.7	43.9	22.4	0.20
randomized	40.7	40.7	22.3	0.13

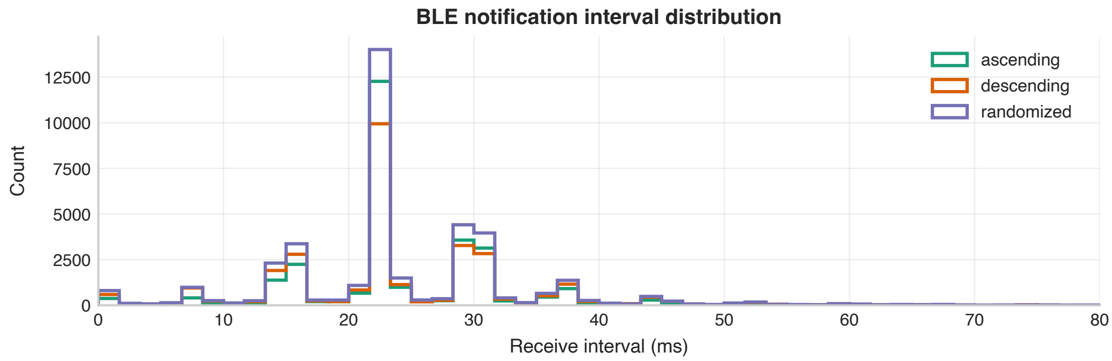


Figure 5: Distribution of directly measured BLE notification intervals during the three orientation runs. The distribution characterizes update timing and jitter, not absolute end-to-end latency.

4.2 Experiment II - Measured Acoustic Reconstruction

The measured BRIR bank supplied the two-channel rendering used in the remaining offline experiments. Figure 6 compares the recorded sweeps with their reconvolved predictions. The captured electrical reference is convolved with the estimated response and compared with the recorded microphone sweep. Same-trial results are an optimistic deconvolution check, whereas cross-repetition results evaluate repeatability between independent sweeps. Across 148 rows, same-trial prediction reached a mean SDR of 25.91 dB, a mean correlation of 0.998, and a Normalized Root Mean Square Error (NRMSE) of 0.051. Cross-repetition prediction, which is the more demanding test, gave a mean SDR of 23.23 dB, a mean correlation of 0.997, and an NRMSE of 0.070, with no appreciable difference between loudspeakers A and B.

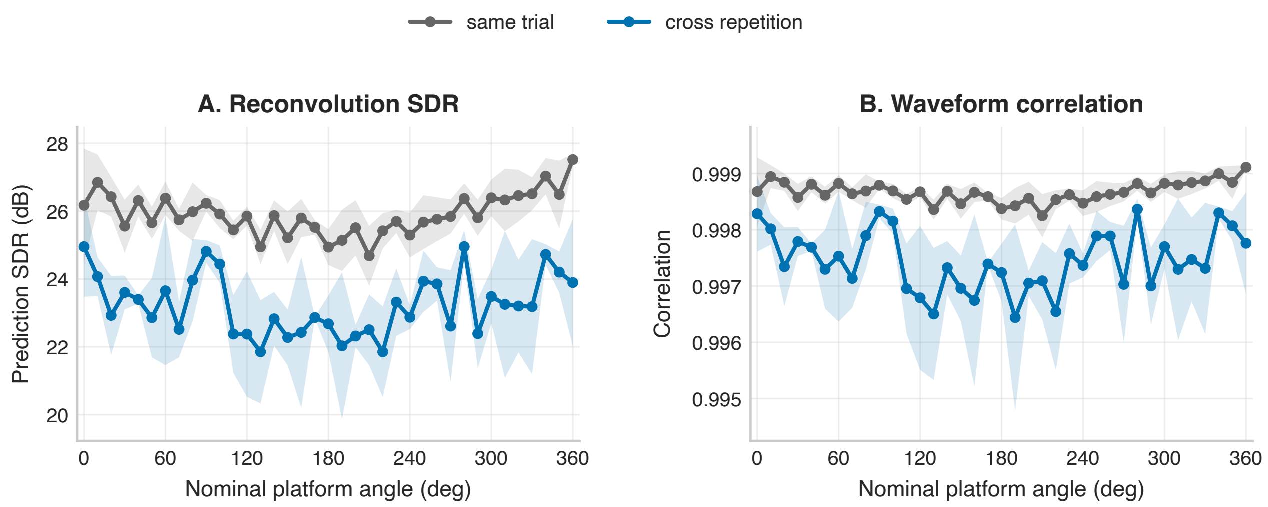


Figure 6: Reconvolution validation of the measured two-channel acoustic responses.

For that reason, the BRIR bank was used in the controlled comparisons of Experiments III through V. The validation does not establish human binaural perception, standardized HRTFs, or ear-canal acoustics, and the remaining mismatch may come from small angular-placement differences, room variation, loudspeaker or microphone noise, and ordinary sweep-to-sweep variability. Thus, the validation supports the intended parameter study without implying perceptually realistic binaural reproduction [14].

4.3 Experiment III - Angular Attention Width

Figure 7 presents the global ranking of the Gaussian widths, where the score is the lower bound of the deterministic 95% bootstrap confidence interval of the target-side angular mean ΔTIR, aggregated over 100 speech pairs, two targets, BRIR repetitions, and channels. According to the target-side angular mean ΔTIR criterion, $\sigma = 30°$ ranked first. Its mean benefit and the lower bound of its 95% bootstrap confidence interval were both 5.47 dB. The mean values for the other widths were 5.08 dB at 45°, 4.77 dB at 20°, 4.11 dB at 60°, and 2.77 dB at 10°, with every speech pair retaining a positive aggregate benefit.

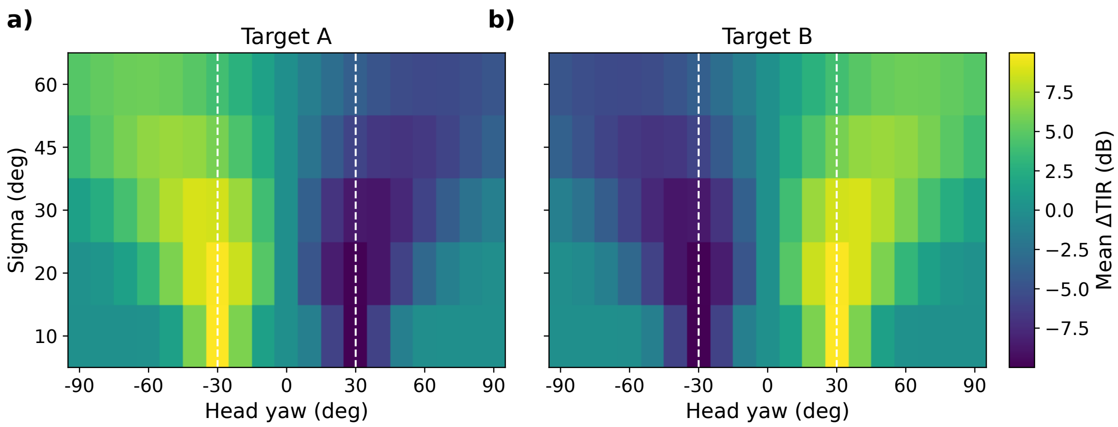


Figure 7: Global sigma ranking from the offline acoustic benchmark.

This result is tied to the benchmark configuration and should not be interpreted as a universal optimum. It depends on the two-source ±30° geometry, the boost-only 10 dB gain law, the measured response bank, and the target-side integration rule. Within that configuration, the intermediate width gave the best overall balance. A very narrow response concentrated the gain over too small an angular interval, whereas a very broad response reduced separation between target and interferer. On that basis, $\sigma = 30°$ was used as the representative width in the separator-delay analysis.

4.4 Experiment IV - Control-Path Latency

Experiment IV varied only the orientation supplied to the controller. As shown in Fig. 8, the acoustic scene follows the undelayed head trajectory, while the attention gains use delayed yaw. Heatmaps report TIR loss relative to zero-delay control and RMS source-gain error as functions of Gaussian width, control delay, and angular velocity. Results are emulated offline using measured BRIRs. Performance depends on the mismatch between the physical or reconstructed acoustic scene and the delayed yaw, rather than on output-audio delay alone. For $\sigma = 30°$, a 20 ms delay produced TIR losses of 0.22, 0.23, and 0.27 dB at 30, 60, and 120 °/s, respectively. With an 80 ms delay, the losses were 0.93, 1.03, and 1.35 dB. At 200 ms, they reached 2.53, 3.01, and 3.82 dB for the same three velocities.

The corresponding angular lag is $\Delta\theta_{\text{lag}} = \omega\tau_c$. With $\tau_c = 200ms$, this gives 6°, 12°, and 24°for 30, 60, and 120 °/s. The largest loss in the tested grid occurred at $\sigma = 20°$, 120 °/s, and 200 ms: 4.44 dB of TIR loss with 2.06 dB RMS gain error. Increasing σ reduced the gain error caused by a given lag, although it also reduced zero-delay selectivity. For the trajectories examined here, delays in the 20–40 ms range

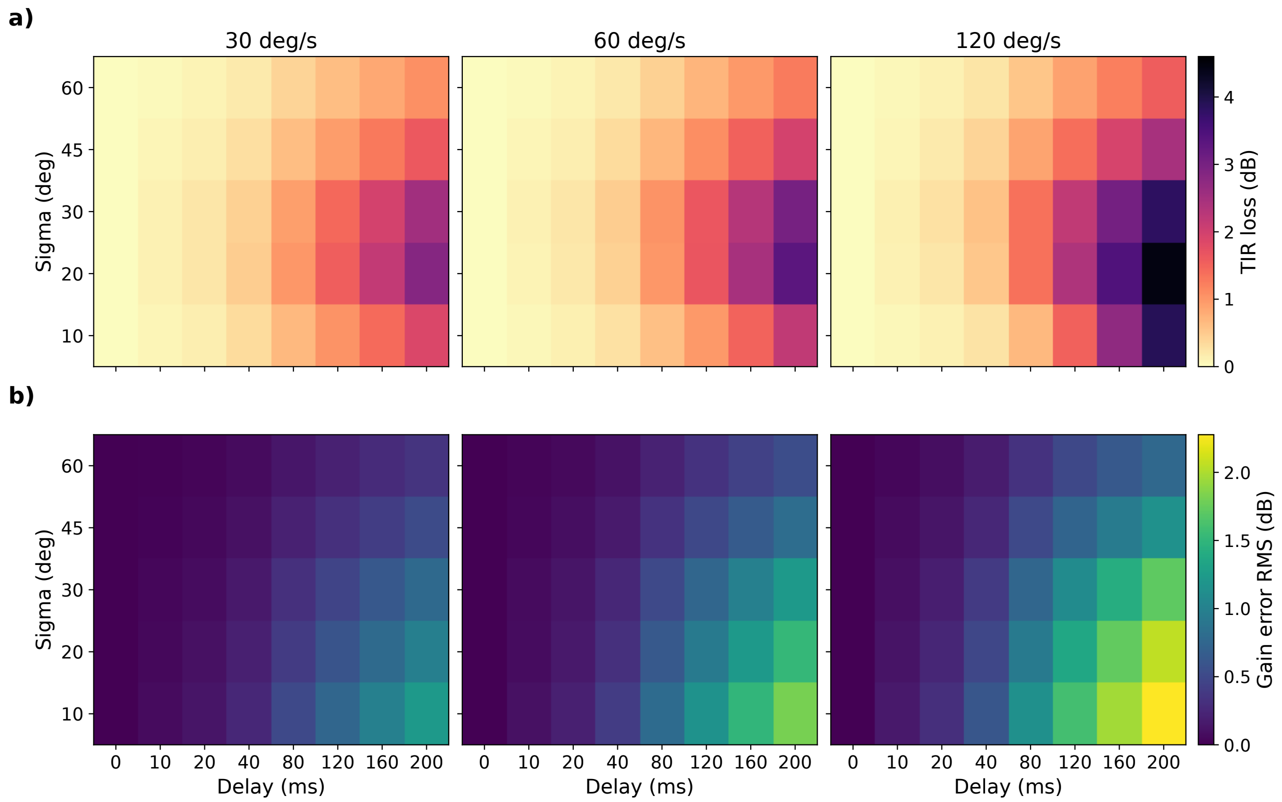


Figure 8: Control-path latency sensitivity analyzed for delays between 0 and 200 ms, and σ between 10°and 60°.

caused relatively small losses, while delays of 160–200 ms materially weakened source-selective control.

4.5 Experiment V - Separator Leakage and Source-Estimate Delay

Figure 9 shows the effect of separator leakage TIR loss relative to the ideal zero-orientation delay, no-leakage separator for different separator SDR settings, sigmas, and angular velocities. Results are emulated after the measured two-channel source-image synthesis. For $\sigma = 30^\circ$ and zero source-estimate delay, the ideal no-leakage condition produced TIR improvements of 5.79, 7.34, and 7.93 dB at 30, 60, and 120 °/s. At a separator SDR of 20 dB, the losses relative to this ideal condition were 1.10, 1.44, and 1.58 dB. At 10 dB, the losses rose to 2.85, 3.67, and 3.99 dB. With a 0 dB separator SDR, the improvement was nearly eliminated in the representative condition.

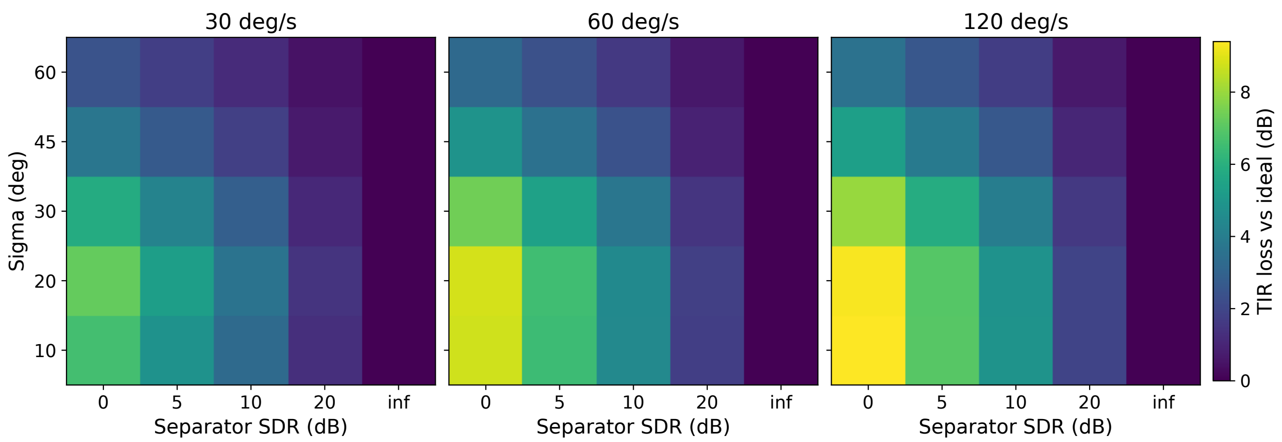


Figure 9: Effect of controlled source-separation leakage over 0 to inf dB SDR and σ between 10°and 60°.

The result follows directly from the object-level signal model. Leakage places residual interferer content inside the object that receives the target-directed gain. A timely controller cannot restore strong selectivity when the nominal target object already contains a substantial copy of the competing source. Separator quality is therefore a primary system constraint rather than a secondary implementation detail.

Source-estimate delay affects the output differently. In this analysis, the output is a live acoustic scene plus delayed source-estimate reinforcement, $y = (x_A + x_B) + (g_A - 1)\hat{x}_A + (g_B - 1)\hat{x}_B$. When both estimated objects are shifted together, finite-window TIR does not always capture the mismatch between the live scene and the delayed reinforcement. For this reason, Fig. 10 reports STOI loss relative to the ideal zero-delay, no-leakage source-overlay output as a function of source-estimate delay, separator SDR, and angular velocity. With $\sigma = 30^\circ$ and an otherwise ideal separator, a 20 ms source delay produced STOI losses of 0.246, 0.266, and 0.276 at 30, 60, and 120 °/s. At 200 ms, the losses increased to 0.521, 0.569, and 0.582. Finite separator SDR caused additional degradation, particularly at short delays where leakage and temporal mismatch are both present.

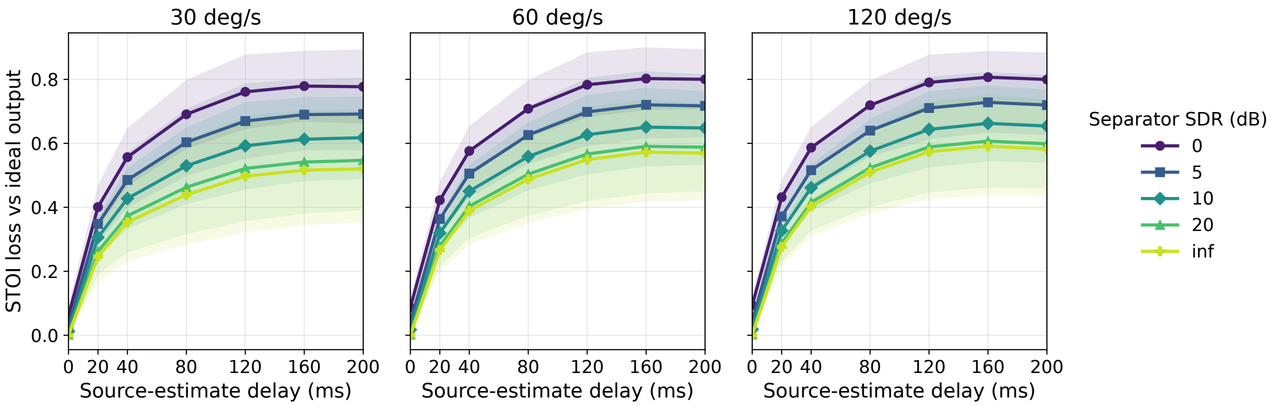


Figure 10: Source-estimate delay impact for $\sigma = 30^\circ$ over the 30°/s, 60°/s and 120°/s ranges. Shaded regions indicate one standard deviation across the 100 LibriSpeech source pairs.

The requirement analysis was done based on the same STOI-loss. Since this analysis used the representative $\sigma = 30^\circ$ condition, it covered 21 velocity–delay combinations and searched over the nominal separator-SDR settings for each one. With a conservative acceptance threshold of 0.05 STOI loss relative to the ideal zero-delay, no-leakage source-overlay output, only the three zero-delay cases were accepted, one for each angular velocity, and the least demanding accepted separator setting was a nominal 5 dB SDR. Relaxing the threshold to 0.10 made the zero-delay cases acceptable even at 0 dB SDR, but did not change the main conclusion: no source-estimate delay of 20 ms or more satisfied the criterion, even under the ideal no-leakage condition. Within this benchmark, source-estimate latency therefore dominated the intelligibility-oriented requirement once the separated source was overlaid on the live acoustic scene.

5. Discussion

The results provide a clearer picture of how a practical head-orientation-driven AAR system should be organized. The orientation measurements showed that the short-term yaw estimate is sufficiently repeatable for controlling a smooth angular gain function, but the accumulated drift prevents the BMI270 from providing a reliable absolute heading over longer periods. The post-hoc residual error was small compared with the selected $\sigma = 30^\circ$ width, whereas the uncorrected error reached tens of degrees. A practical implementation therefore requires an absolute orientation anchor. For Tiresias, this could be provided by adding a magnetometer or by combining the IMU with another periodic heading reference.

The attention-control path was comparatively tolerant to latency. For $\sigma = 30^\circ$, a 20 ms control delay caused less than 0.3 dB of TIR loss at all tested angular velocities, while delays of 80 ms remained close to or below 1.35 dB. Larger degradation appeared mainly for delays above 120–160 ms and for faster head motion. The measured BLE notification rate of 40.7 Hz corresponds to an average update spacing of approximately 24.6 ms, with P95 intervals between 37.5 and 43.9 ms. These values lie within the low-loss region of the offline control-delay analysis, suggesting that the BLE update granularity is suitable for this form of gain control. This comparison concerns update spacing rather than absolute motion-to-host latency, which must still be measured with synchronized timing in a real-time implementation.

The attention-width results also show that σ should be selected from the spatial arrangement of the sources rather than treated as a fixed parameter. With sources at $\pm 30^{\circ}$, the intermediate value $\sigma = 30^{\circ}$ provided the largest target-side mean ΔTIR. Narrower patterns produced stronger gain concentration but covered a smaller useful angular region and were more sensitive to orientation mismatch. Wider patterns tolerated head motion but applied increasingly similar gains to both sources. In a practical system, the attention width could therefore be adapted according to the angular separation between detected source objects, using narrower patterns for well-separated sources and broader patterns when source positions are uncertain or closely spaced.

The measured BRIR bank made it possible to evaluate these effects while retaining the acoustic differences introduced by the room, mannequin, microphones, and loudspeaker geometry. Its cross-repetition agreement indicates that the measurement procedure was consistent enough for comparative testing. More importantly, the experiments separate the requirements of the control path from those of the audio-object path. Orientation update delay produced gradual degradation, leaving a usable timing margin for sensing, transmission, and gain computation. The separator path was substantially more restrictive. Cross-source leakage directly reduced the benefit of object-level gain control because residual interference received the same gain as the selected object. Source-estimate delay was even more critical in the live-scene-plus-reinforcement configuration, where a delay of 20 ms already exceeded the adopted STOI-loss criterion, including under ideal no-leakage conditions.

These results indicate that the main implementation challenge is not the Gaussian attention controller itself. A practical system can tolerate moderate orientation-update intervals, provided that long-term heading drift is corrected and that the attention width is matched to the source geometry. The critical path is the generation of source objects with sufficiently low latency and leakage. Future development should therefore prioritize a low-latency real-time separator, synchronized source rendering, and an absolute-heading reference on the Tiresias platform before moving to listener-based evaluation. The present objective results do not establish perceptual or clinical benefit, but they define a concrete set of subsystem requirements for implementing and testing the complete system.

6. Conclusion

This study quantified the main engineering constraints of a head-orientation-driven AAR speech-enhancement pipeline, namely orientation telemetry, measured acoustic reconstruction, Gaussian attention width, control-path delay, separator leakage, and source-estimate delay. Raw yaw drift ranged from 2.44 to 3.23°/min, and after offline drift correction the Tiresias yaw measurements had a residual RMSE of 0.47°. BLE notifications arrived at an average rate of 40.7 Hz. The measured two-channel response bank predicted independent sweeps with a cross-repetition SDR of 23.23 dB, supporting its use in the offline benchmarks. For the tested $\pm 30^{\circ}$ geometry, $\sigma = 30^{\circ}$ gave the highest aggregate angular TIR score. Under faster motion, control delays above about 80 ms produced multi-dB losses, and separator leakage reduced much of the ideal object-level advantage. Source-estimate delay also lowered STOI relative to the ideal source-overlay output. The evaluated framework is therefore technically feasible only within a bounded operating region in which yaw drift, control latency, leakage, and source delay are actively managed. Perceptual and clinical claims will require subsequent studies with listeners, including hearing-loss-specific validation.

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